ET/ME/M/HND/21/002

(i)

Zakes 17/4/23

Manual Part Programming.

Introduction:

Types of part programming, Computer aided part programming, part programming manual, part ~~programme~~ using Sub routines, do loops and fixed Cycles are described as explained below. ~~the chapter~~.

Types of part programming.

The part program is a sequence of instructions, which describe the work, which has to be done on a part, in the form required by a computer under the control of a numerical control computer program. It is the task of preparing a ~~program~~ sheet from a drawing sheet. All data is fed into the numerical control system using a standardized format. Programming is where all the machining data are compiled and where the data are translated into a language which can be understood by the control system of the machine tool. The machining data is as follows:

(i) Machining sequence classification of process, tool start up point, cutting depth, tool part.

(ii) Cutting conditions, Spindle speed, feed rate, coolant.

(iii) Selection of cutting tools.

Whereas preparing a part program, need to perform the following steps:

(i) Determine the start up procedure, which includes the extraction of dimensional data from part drawings and data regarding surface quality requirements on the machined component.

(ii) Select the tool and determine the tool offset.

(iii) Set up the zero position for the workpiece.

(iv) Select the speed and rotation of ~~the~~ Spindle.

(v) Set up the tool motions according ~~to~~ the profile required.

(vi) Return the cutting tool to the reference point after completion of work.

(vii) End the program by stopping the spindle and coolant.

The part programming contains the list of coordinate values along ~~the~~ X, Y and Z directions of the entire tool path to finish the component. The program should ~~also~~ contain information, such as feed and speed. Each of the necessary instruction for a particular operation given in the part program is known as an NC word. A group of such NC words constitutes a complete NC instruction, known as block. Hence the methods of part programming can be of two types:

depending upon the two techniques as below:
(a) Manual part programming and (b) Computer aided part programming.

Manual part programming.

The programmer first prepares the program manuscript in a standard format. Manuscripts are typed with a device known as flexo writer, which is also used to type the program instructions. After the program is typed, the punched tape is prepared on the flexo writer. Complex shaped components require tedious calculations. This type of programming is carried out for simple machining parts produced on point-to-point machine tool. To be able to create a part program manually, need the following information.

(i) Knowledge about various manufacturing processes and machines.
(ii) Sequence of the operation to be performed for a given component.
(iii) Knowledge of the selection of cutting parameters.
(iv) Editing the part program according to the design changes.
(v) Knowledge about the codes and functions used in part programs.

Elements and Structure.

The most common elements of the narrative structure are setting, plot and theme. The parts of narrative plot include exposition, rising action, climax, falling action and resolution. The resolution is also called the denouement.

Structural element are used in structural analysis to split a complex structure into simple elements. Within a structure, an element cannot be broken down (decomposed) into parts of different kinds (e.g beam or column, shaft etc). Structural element can be lines, surfaces ~~or~~ volumes.

Tooling Information

Tooling, also known as machine tooling, is the process of acquiring the manufacturing components and machines needed for production. The common categories of machine tooling include fixtures, jigs, gauges, molds, dies, cutting equipment and patterns.

A tooling engineer uses computer-aided (CAD) manufacturing to create and adjust tools or parts for vehicles, airplanes, heavy equipment or any product that needs customized components.

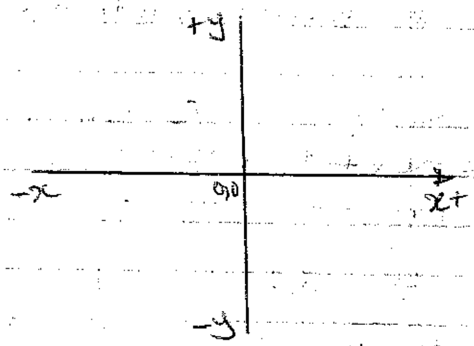
Computer skills are just also important because tooling engineers build and troubleshoot design with CAD software.

Cartesian Coordinate System,

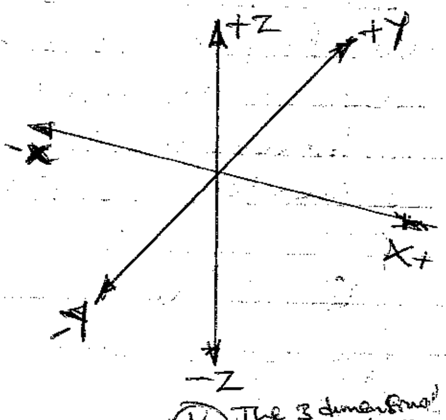
Almost everything can be produced on a conventional machine it can be produced on a computer numerical control machine tool, with its many advantages. The machine tool movements used in producing a product are two basic types: point-to-point (straight-line movements) and continuous path (contouring movements)

The cartesian or rectangular, coordinat system was devised by French mathematician and philosopher Rene' Descartes. With this system, any specific point can be described in mathematical terms from any other point along three perpendicular axes. This concept fits machine tools perfectly since their contruction is generally based on three axes of motion (X, Y, Z) plus an axis of rotation. On a plain vertical milling machine, the X axis is the horizontal movement (right or left) of the table, the Y axis is the table cross movement (toward or away from the column) and the Z axis is the vertical movement of the knee or the spindle. CNC systems rely heavily on the use of rectangular coordinates because the programmer can locate every point on a job precisely.

When point are located on a workpiece, the straight intersecting lines, one vertical and one horizontal are used. These lines must be at right angles to each other, and the point where they cross is called the origin or zero point as shown in the diagram ~~below~~.

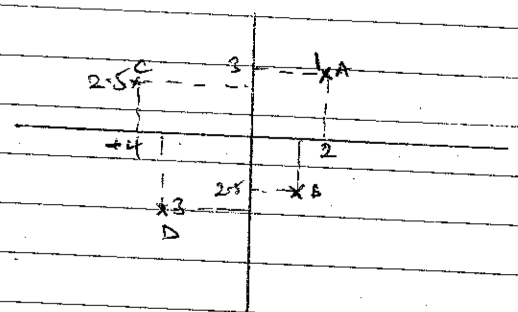


(a) Intersecting lines form angles and establish the zero point



(b) The 3 dimensional coordinate planes axes used in CNC

The three dimensional coordinate planes are shown in diagram (b). The X and Y planes (axes) are horizontal and represent horizontal machine table motions. The Z plane or axis represents the vertical tool motion. The plus (+) and minus (-) signs indicate the direction from the zero point (origin) along the axis of movement. The four quadrant formed when the XY axes cross are numbered in a counterclockwise direction (fig. c). All positions located in quadrant 1 would be positive (+X) and positive (+Y) e.t.c.



In diagram above, point A (2,1), B (1.5, -2.5), C (-4, 2.5) and D (-3.75, -3).

Absolute and incremental Coordinates.

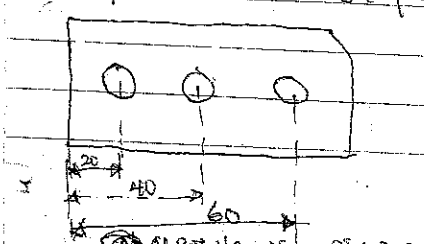
~~ISO Standard~~

~~Absolute~~ There are two different ways to use ~~both~~ cartesian and polar coordinates. How we use them depends on whether the coordinate refers to the origin (more common) or takes its reference from the previous point called incremental coordinate.

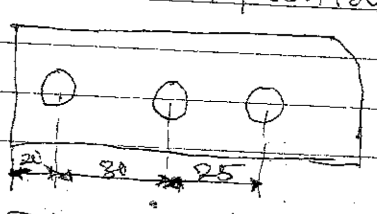
NB: (i) Coordinate values of command for moving the tool can be indicated by absolute or incremental designation as shown in diagram below.

(ii) Absolute command: The tool moves to a point at "the distance from zero point of the coordinate system" that is to the position of the coordinate values.

(iii) Incremental commands: Specify the distance from the previous tool position to the next tool position.



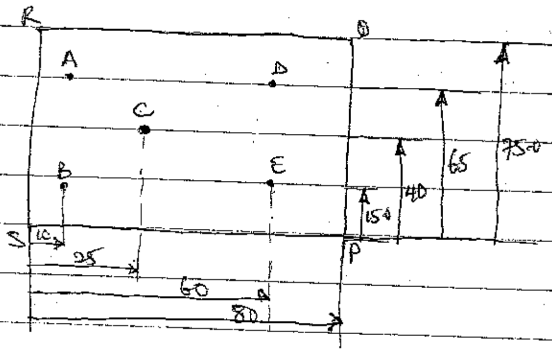
(A) Absolute Dimensioning



(B) Incremental Dimensioning

(3)

Example ① Using edge S as origin in figure below determine the coordinates of A, B, C, ~~D~~ and E.



Solution

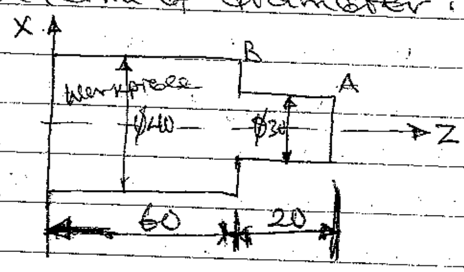
Position	X	Y
A	10	65
B	10	15
C	25	40
D	60	65
E	60	15

Example ② Using the same diagram above and use the Q as origin, determine the coordinate of A, B, C, D and E

Solution

Position	X	Y
A	−70	−10
B	−70	−60
C	−55	−35
D	−20	−10
E	−20	−60

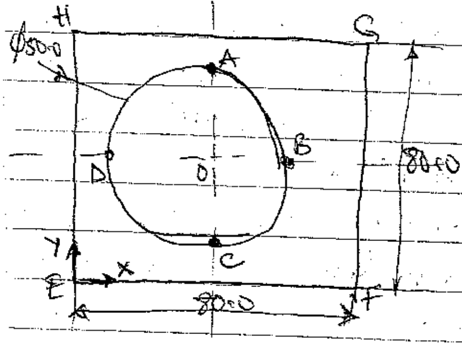
Example 3: Determine the coordinate values of A and B in the diagram below in term of diameter.



Solution

Position	X	Z
A	30.0	80.0
B	40	60.0

Example 4: Using Centre O as origin in diagram below, determine the coordinates of A, B, C and D



Sol

Position	X	Y
A	0	25
B	25	0
C	0	-25
D	-25	0

ISO Standards.

ISO is the international organization for standardization. It has a membership of 161 national standard bodies from countries large and small, industrialized, developing and in transition, in all regions of the world. ISO standard objective is to provide confidence when a product or service meets the specifications or requirements of an ISO standard, this provides confidence that they incorporate essential features. These features can include quality, safety. ISO standard also help to ensure such benefits at an economical cost.

Machine codes for machine auxiliary function and Movement

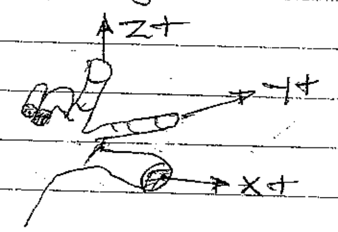
CNC machines are very accurate and powerful industrial robots developed jointly by Mr. John Parsons, IBM and Massachusetts Institute of Technology Servo-mechanism Laboratory in the 1950's. Most CNC Machine tools use a language set by the Electronic Industry Association (EIA) in the 1960's. The official name of this language is RS-274D, but everyone refers it "G-Code" or "G & M Code" because many of the words of this language begin with the letters G or M.

While many of the words used by different CNC machines are the same, there are differences between makes and models. This is due in part to machines having different configurations and options. For examples, a machine with a chip conveyor will have words to turn the conveyor on and off, while a machine without a conveyor does not. So, while RS-274D is a standard, it is not rigid or enforced. Always refer to the machine documentation for the exact words and syntax for your CNC machine.

Standard G and M Codes.

G and M codes are special programming language that is interpreted by Computer Numerical Control (CNC) machines to create motion and other tasks.

It is a language that can be quite complex at times and can vary from machine to machine. The basics, however, are much simpler than it first appears and for the most part follows an industry adopted standard.
An important point to remember when reading this manual: In describing motion of a machine it will always be described as tool movement relative to the work piece. In many machines the work piece will move in more axes than the tool; however, the program will always define tool movement around the work piece. Axes direction follows the right hand rule as shown below.



The most common codes use when programming CNC machines tools are G-Codes (preparatory functions) and M codes (miscellaneous functions). Other codes such as F, S, D and T are used for machine functions such as feed, speed, cutter diameter offset, tool number etc. G-codes are sometime called cycle codes because they refer to some action occurring on the X, Y and/or Z-axes of a machine tool. The G-Codes are grouped into categories such as Group 01 containing codes G00, G01, G02, G03 which cause some movement of the machine table or head.
Group 03 include either absolute or incremental programming. A G00 code rapidly positions the cutting tool while it is above the workpiece from one point to another point on a job. During the rapid traverse movement, either the X or Y-axes can be moved individually or both axes can be moved at the same time. The rate of rapid travel varies from machine to machine.
NB: (1) G-code is a programming language for CNC that instructs machines where and how to move. Most machines

Speak a different "dialect" of G-Code, so the codes vary depending on type, make and model. Each machine comes with an instruction manual that shows that particular machines codes for a specific function.

(ii) M-Code is an ~~[illegible]~~ auxiliary command; description vary - Many M-code call for machine functions like "open workstation door" which is why some say M stand for machine through it was not intended ~~to~~ or it used by the CNC to command on/off signals to the machine function i.e M03 — Spindle forward (CW), M05 - Spindle stop etc.

(iii) A G - Codes Command are written as

GKK X Y Z

Traverse code — Coordinate values

The total numbers of these codes are 100, out of which some of important codes are given as under with their functions:

G - Codes (preparatory function) & M codes (mis.

Continue on page 5 ~~[illegible]~~

~~Example~~ : The component shown in diagram below is to be machined out of 20mm diameter material, using C-502 CNC lathe machine. Use the G&M-code language to write a complete turning program according to the drawing. Use 2000 rev/min as spindle speed, 150mm/min as ~~feed~~ and 2mm as depth of cut at a time -